Experiment No. 15 – Harris Corner Detection

1. Aim

To perform basic image handling and processing operations by reading an image in Python and detecting the corners in the image using the Harris Corner Detection function.

2. Program

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Read the input image

image = cv2.imread('/content/sample.jpg')


# Convert image into grayscale

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


# Convert image to float32

gray = np.float32(gray)


# Apply Harris Corner Detection

corners = cv2.cornerHarris(gray, 2, 3, 0.04)


# Dilate the detected corners

corners = cv2.dilate(corners, None)


# Mark detected corners in the original image
```

```
image[corners > 0.01 * corners.max()] = [0, 0, 255]


# Display the output

plt.figure(figsize=(8, 6))

plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))

plt.title('Harris Corner Detection')

plt.axis('off')

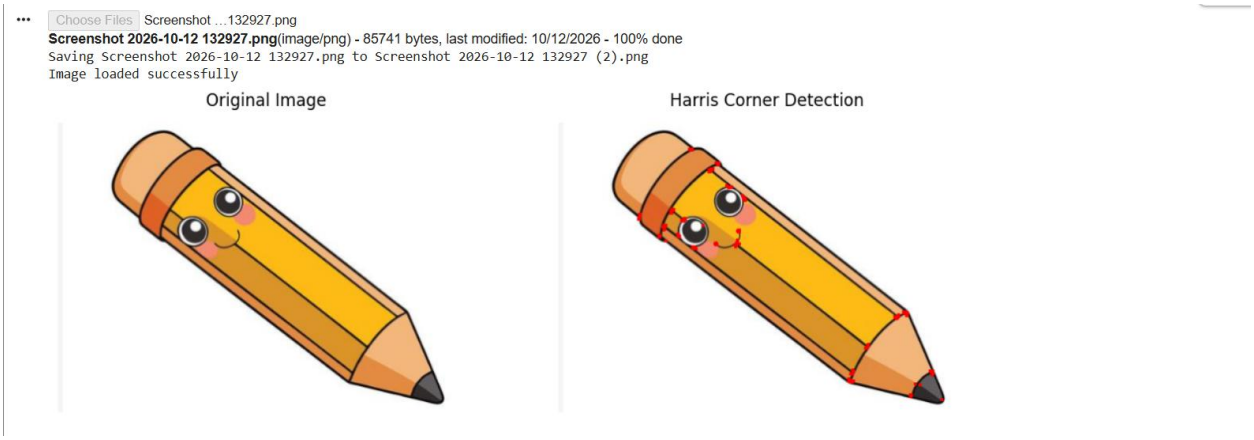
plt.show()
```

3. Expected Output

The output image should show:

- Original image with detected corners.
- Red markings indicating the detected corner points.
- Corners detected at locations where two edges meet.

4. Output



The Harris Corner Detection algorithm successfully identifies and highlights the corners in the given image using red markings.

5. Result

Thus, the image was successfully read and processed, and the corners were detected using the Harris Corner Detection function in Python and OpenCV.